

	$= 200 \text{ W}$	answer
2(a)	Gravitational potential at a point is the <u>work done per unit mass</u> in <u>bringing a small test mass from infinity to that point</u> .	[1]
2(b)(i)	Potential at infinity is taken to be zero. Due to <u>the attractive nature of the gravitational force</u>, the <u>direction of external force is opposite to the direction of displacement of the mass</u>, hence <u>work done by an external force</u> to bring any mass from infinity to that point is <u>always negative</u>	[1] for attractive nature of gravitational force [1] opposite direction between external force and displacement and hence negative work done.
2(b)(ii)	$\phi = -\frac{GM}{r}$ $-2.32 \times 10^8 = -(6.67 \times 10^{-11})M(3.95 \times 10^{-8})$ $M = 8.806 \times 10^{25} \approx 8.81 \times 10^{25} \text{ kg}$	[1] for correct substitution [1] for correct answer
2(b)(iii)	Gravitational force on the satellite provides the centripetal force for it to move in circular path. $F_g = F_c$ $\frac{GMm}{r^2} = mr\omega^2$ $GM = r^3\omega^2$ $(6.67 \times 10^{-11} \times 8.806 \times 10^{25}) = r^3 \left(\frac{2\pi}{0.72 \times 24 \times 60 \times 60} \right)^2$ $r = 8.319 \times 10^7 \approx 8.3 \times 10^7 \text{ m}$	[1] for correct statement [1] for correct expression [1] for correct answer
2(b)(iv)	total energy = KE + GPE $= \frac{1}{2}mv^2 + \left(-\frac{GMm}{r} \right)$ $= \frac{1}{2}(1200)(8400)^2 + \left(-\frac{6.67 \times 10^{-11} \times 8.806 \times 10^{25} \times 1200}{8.319 \times 10^7} \right)$ $= -4.239 \times 10^{10} \text{ J}$ At infinity, total energy of the satellite is zero Additional energy to be supplied = $4.239 \times 10^{10} \approx 4.2 \times 10^{10} \text{ J}$	[1] for correct substitution [1] for correct answer for total energy [1] for correct answer (accept answer without explanation)

3

No.	Solution	Remarks